3 **(a)** State the first law of thermodynamics.

....

.... [2]

(b) Some neon gas, assumed to behave ideally, is contained within a cylinder which is insulated to prevent loss of heat as shown in Fig. 3.1.

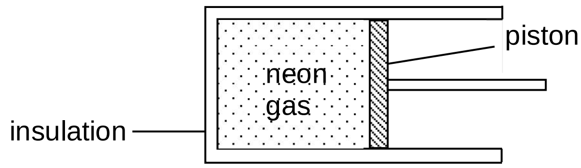

Fig. 3.1

Initially the volume of neon gas is 2.90×10^{-4} m^3, its pressure is 1.04×10^5 Pa and its temperature is 314 K.

(i) The neon gas is then compressed to a volume of 2.90×10^{-5} m^3 and its temperature rises to 790 K. Calculate the pressure of the neon gas after this compression.

pressure = ... Pa [2]

(ii) The work done on the neon gas during the compression is 90 J. Calculate the change in the internal energy of the neon gas during the compression.

change in internal energy = ... J [1]

(iii) Explain the meaning of internal energy as applied to this system and use your result in **(ii)** to explain why the temperature of the neon gas increases during the compression.

....

....

....

....

.... [2]

(c) Another cylinder identical to that in Fig. 3.1 is divided into two sections by a thermally-insulating partition. One section contains 3.00 mol of neon gas at 400 K, and the other 1.00 mol of helium gas at 600 K. The partition is then removed.

(i) Calculate the equilibrium temperature of the mixture of gases. Both neon and helium are monatomic ideal gases.

equilibrium temperature = ... K [2]

(ii) If the root-mean-square speed of the neon atoms is v at this equilibrium temperature, express the root-mean-square speed of the helium atoms in terms of v. The molar masses of helium and neon are 4 $g\,mol^{-1}$ and 20 $g\,mol^{-1}$ respectively.

root-mean-square speed = ... [2]

4 **(a)** State the conditions for the formation of a stationary wave.